Алгоритм МСЕ 3D-пружності: повний ланцюжок формул

Теги праворуч: `[файл.cs:рядок]` $\quad \varphi_{ix} \equiv \frac{\partial \varphi_i}{\partial x} \quad \xi_0{=}\alpha,\ \xi_1{=}\beta,\ \xi_2{=}\gamma$

1 — Матеріальні параметри

$$\lambda = \frac{E}{(1+\nu)(1-2\nu)}, \qquad \mu = \frac{E}{2(1+\nu)} \qquad \texttt{[Program.cs:175-176]}$$

$$\lambda_s = \frac{\lambda - 2\mu}{2} = \frac{E\nu}{(1+\nu)(1-2\nu)} \qquad \texttt{[ElementStiffness.cs:128]}$$

$$C_{1111} = \lambda_s + 2\mu = \lambda(1-\nu), \qquad C_{1122} = \lambda_s = \lambda\nu \qquad \texttt{[ElementStiffness.cs:129-130]}$$

Приклад: сталь $E{=}2.1{\times}10^{11}$ Па, $\nu{=}0.3$:

$$\lambda = \frac{2.1{\times}10^{11}}{1.3{\cdot}0.4} = 4.04{\times}10^{11}, \quad \mu = \frac{2.1{\times}10^{11}}{2.6} = 8.08{\times}10^{10}, \quad C_{1111} = 5.65{\times}10^{11}, \quad C_{1122} = 1.21{\times}10^{11}\ \text{Па}$$

2 — Генерація сітки

$$\text{вузол } (i_x, i_y, i_z) \text{ існує } \Leftrightarrow\ (i_x \bmod 2) + (i_y \bmod 2) + (i_z \bmod 2) \leqslant 1 \qquad \texttt{[Mesh.cs:89-91]}$$

$$x_n = i_x \cdot \tfrac{A_x}{2N_x}, \quad y_n = i_y \cdot \tfrac{A_y}{2N_y}, \quad z_n = i_z \cdot \tfrac{A_z}{2N_z} \qquad \texttt{[Mesh.cs:115-117]}$$

$$e = e_z N_y N_x + e_y N_x + e_x, \qquad \mathrm{NT}[e,i] = \mathrm{NodeIndex}[2e_z{+}\ldots,\ 2e_y{+}\ldots,\ 2e_x{+}\ldots] \qquad \texttt{[Mesh.cs:141-169]}$$

3 — Квадратура Гауса (3 точки)

$$\xi_k \in \{-\sqrt{3/5},\ 0,\ \sqrt{3/5}\}, \qquad w_k \in \{5/9,\ 8/9,\ 5/9\} \qquad \texttt{[ShapeFunctions.cs:46-54]}$$

$$\int_{-1}^{1} f\,d\xi \approx \sum_{k=0}^{2} w_k f(\xi_k), \qquad w = w_{k_1} w_{k_2} w_{k_3}, \quad \mathrm{gp} = 9k_1{+}3k_2{+}k_3 \qquad \texttt{[ElementStiffness.cs:72-79]}$$

4 — Локальні координати 20 вузлів серендіп-елемента

`[ShapeFunctions.cs:18-41]`

i	α_i	β_i	γ_i	тип	i	α_i	β_i	γ_i	тип
0	−1	−1	−1	кут	10	0	+1	−1	$\alpha{=}0$
1	+1	−1	−1	кут	11	−1	0	−1	$\beta{=}0$
2	+1	+1	−1	кут	12	0	−1	+1	$\alpha{=}0$
3	−1	+1	−1	кут	13	+1	0	+1	$\beta{=}0$
4	−1	−1	+1	кут	14	0	+1	+1	$\alpha{=}0$
5	+1	−1	+1	кут	15	−1	0	+1	$\beta{=}0$
6	+1	+1	+1	кут	16	−1	−1	0	$\gamma{=}0$
7	−1	+1	+1	кут	17	+1	−1	0	$\gamma{=}0$
8	0	−1	−1	$\alpha{=}0$	18	+1	+1	0	$\gamma{=}0$
9	+1	0	−1	$\beta{=}0$	19	−1	+1	0	$\gamma{=}0$

5 — Функції форми

$$\varphi_i \text{ (кут, i=0..7)} = \tfrac{1}{8}(1+\alpha\alpha_i)(1+\beta\beta_i)(1+\gamma\gamma_i)(\alpha\alpha_i + \beta\beta_i + \gamma\gamma_i - 2) \qquad \texttt{[ShapeFunctions.cs:74]}$$

$$\varphi_i\ (\alpha_i{=}0) = \tfrac{1}{4}(1-\alpha^2)(1+\beta\beta_i)(1+\gamma\gamma_i) \qquad \texttt{[ShapeFunctions.cs:85]}$$

$$\varphi_i\ (\beta_i{=}0) = \tfrac{1}{4}(1+\alpha\alpha_i)(1-\beta^2)(1+\gamma\gamma_i) \qquad \texttt{[ShapeFunctions.cs:87]}$$

$$\varphi_i\ (\gamma_i{=}0) = \tfrac{1}{4}(1+\alpha\alpha_i)(1+\beta\beta_i)(1-\gamma^2) \qquad \texttt{[ShapeFunctions.cs:89]}$$

Приклад: вузол 0 $(\alpha_0{=}\beta_0{=}\gamma_0{=}-1)$:

$$\varphi_0 = \tfrac{1}{8}(1-\alpha)(1-\beta)(1-\gamma)(-\alpha-\beta-\gamma-2), \qquad \varphi_0(-1,-1,-1) = 1, \quad \varphi_0(+1,\cdot,\cdot) = 0$$

Похідні — кутові вузли $(i = 0..7)$:

$$\frac{\partial \varphi_i}{\partial \alpha} = \tfrac{\alpha_i}{8}(1+\beta\beta_i)(1+\gamma\gamma_i)(2\alpha\alpha_i + \beta\beta_i + \gamma\gamma_i - 1) \qquad \texttt{[ShapeFunctions.cs:110]}$$

$$\frac{\partial \varphi_i}{\partial \beta} = \tfrac{\beta_i}{8}(1+\alpha\alpha_i)(1+\gamma\gamma_i)(\alpha\alpha_i + 2\beta\beta_i + \gamma\gamma_i - 1) \qquad \texttt{[ShapeFunctions.cs:113]}$$

$$\frac{\partial \varphi_i}{\partial \gamma} = \tfrac{\gamma_i}{8}(1+\alpha\alpha_i)(1+\beta\beta_i)(\alpha\alpha_i + \beta\beta_i + 2\gamma\gamma_i - 1) \qquad \texttt{[ShapeFunctions.cs:115]}$$

Похідні — серединні (α_i=0):

$\frac{\partial\varphi_i}{\partial\alpha} = -\frac{\alpha}{2}(1+\beta\beta_i)(1+\gamma\gamma_i),\ \frac{\partial\varphi_i}{\partial\beta} = \frac{\beta_i}{4}(1-\alpha^2)(1+\gamma\gamma_i),\ \frac{\partial\varphi_i}{\partial\gamma} = \frac{\gamma_i}{4}(1-\alpha^2)(1+\beta\beta_i)$ `[ShapeFunctions.cs:128-130]`

Похідні — серединні (β_i=0):

$\frac{\partial\varphi_i}{\partial\alpha} = \frac{\alpha_i}{4}(1-\beta^2)(1+\gamma\gamma_i),\ \frac{\partial\varphi_i}{\partial\beta} = -\frac{\beta}{2}(1+\alpha\alpha_i)(1+\gamma\gamma_i),\ \frac{\partial\varphi_i}{\partial\gamma} = \frac{\gamma_i}{4}(1+\alpha\alpha_i)(1-\beta^2)$ `[ShapeFunctions.cs:135-137]`

Похідні — серединні (γ_i=0):

$\frac{\partial\varphi_i}{\partial\alpha} = \frac{\alpha_i}{4}(1+\beta\beta_i)(1-\gamma^2),\ \frac{\partial\varphi_i}{\partial\beta} = \frac{\beta_i}{4}(1+\alpha\alpha_i)(1-\gamma^2),\ \frac{\partial\varphi_i}{\partial\gamma} = -\frac{\gamma}{2}(1+\alpha\alpha_i)(1+\beta\beta_i)$ `[ShapeFunctions.cs:142-144]`

6 — Варіаційна постановка → слабка форма → СЛАР

Сильна форма (індексна нотація; $\partial_j \equiv \partial/\partial x_j$):

$$-\partial_j\sigma_{ij}(\mathbf{u}) = f_i \ \text{ в } \Omega, \qquad u_i = 0 \ \text{ на } \Gamma_D, \qquad \sigma_{ij}n_j = t_i \ \text{ на } \Gamma_N$$

$$\Downarrow \quad \times v_i,\ \sum_i,\ \int_\Omega d\Omega$$

$$-\int_\Omega (\partial_j\sigma_{ij})\, v_i \, d\Omega = \int_\Omega f_i v_i \, d\Omega$$

$$\Downarrow \quad \partial_j(\sigma_{ij}v_i) = (\partial_j\sigma_{ij})v_i + \sigma_{ij}(\partial_j v_i) \implies -(\partial_j\sigma_{ij})v_i = \sigma_{ij}(\partial_j v_i) - \partial_j(\sigma_{ij}v_i)$$

$$\int_\Omega \sigma_{ij}\,\partial_j v_i \, d\Omega \ - \underbrace{\int_\Omega \partial_j(\sigma_{ij}v_i)\, d\Omega}_{=\int_\Gamma \sigma_{ij}n_j v_i\, d\Gamma \text{ (Гаусс–Остроградський)}} = \int_\Omega f_i v_i \, d\Omega$$

$$\Downarrow \quad \int_\Gamma \sigma_{ij}n_j v_i \, d\Gamma = \underbrace{\int_{\Gamma_N} t_i v_i \, d\Gamma}_{\text{ГУ Нейман}} + \underbrace{\int_{\Gamma_D} \sigma_{ij}n_j \cdot 0 \, d\Gamma}_{0,\ v_i=0 \text{ на } \Gamma_D}$$

$$\Downarrow \quad \sigma_{ij} = \sigma_{ji} \ \Rightarrow\ \sigma_{ij}\partial_j v_i = \sigma_{ij}\cdot\tfrac{1}{2}(\partial_j v_i + \partial_i v_j) = \sigma_{ij}\,\varepsilon_{ij}(\mathbf{v}), \quad \text{де } \varepsilon_{ij}(\mathbf{v}) = \tfrac{1}{2}(\partial_j v_i + \partial_i v_j)$$

Слабка форма:

$$\int_\Omega \sigma_{ij}(\mathbf{u})\,\varepsilon_{ij}(\mathbf{v})\, d\Omega = \int_{\Gamma_N} t_i v_i \, d\Gamma + \int_\Omega f_i v_i \, d\Omega$$

$$\Downarrow \quad \text{розписати } \sigma_{ij}\varepsilon_{ij}(\mathbf{v}) \text{ через 6 компонент (закон Гука 3D)}$$

$$\int_\Omega \sigma_{ij}\varepsilon_{ij}(\mathbf{v})\, d\Omega = \int_\Omega \big[\sigma_{xx}\varepsilon_{xx}(\mathbf{v}) + \sigma_{yy}\varepsilon_{yy}(\mathbf{v}) + \sigma_{zz}\varepsilon_{zz}(\mathbf{v}) + \sigma_{xy}\varepsilon_{xy}(\mathbf{v}) + \sigma_{yz}\varepsilon_{yz}(\mathbf{v}) + \sigma_{xz}\varepsilon_{xz}(\mathbf{v})\big]\, d\Omega$$

$$\Downarrow \quad \boldsymbol{\sigma} = C\boldsymbol{\varepsilon}(\mathbf{u}): \quad \sigma_{xx} = C_{1111}\varepsilon_{xx} + C_{1122}(\varepsilon_{yy}+\varepsilon_{zz}),\ \sigma_{xy} = \mu\varepsilon_{xy}, \text{ тощо}$$

$$\underbrace{\int_\Omega \big[C_{1111}(\varepsilon^u_{xx}\varepsilon^v_{xx} + \varepsilon^u_{yy}\varepsilon^v_{yy} + \varepsilon^u_{zz}\varepsilon^v_{zz}) + C_{1122}(\varepsilon^u_{xx}(\varepsilon^v_{yy}+\varepsilon^v_{zz}) + \ldots) + \mu(\varepsilon^u_{xy}\varepsilon^v_{xy} + \varepsilon^u_{yz}\varepsilon^v_{yz} + \varepsilon^u_{xz}\varepsilon^v_{xz})\big]\, d\Omega}_{a(\mathbf{u},\mathbf{v})} = \ell(\mathbf{v})$$

$$\Downarrow \quad u_x = \sum_a \varphi_a u^a_x,\ u_y = \sum_a \varphi_a u^a_y,\ u_z = \sum_a \varphi_a u^a_z \quad \text{(FEM, Гальоркін)}$$

Тест-функція $\mathbf{v} = \varphi_b\mathbf{e}_x$: $\varepsilon^v_{xx} = \frac{\partial\varphi_b}{\partial x},\ \varepsilon^v_{xy} = \frac{\partial\varphi_b}{\partial y},\ \varepsilon^v_{xz} = \frac{\partial\varphi_b}{\partial z},$ всі решта $= 0$.

Рядок СЛАР (b=0..19, компонента x):

$$\sum_a \Big[C_{1111}\int_\Omega \tfrac{\partial\varphi_b}{\partial x}\tfrac{\partial\varphi_a}{\partial x}\, d\Omega + \mu\int_\Omega \tfrac{\partial\varphi_b}{\partial y}\tfrac{\partial\varphi_a}{\partial y}\, d\Omega + \mu\int_\Omega \tfrac{\partial\varphi_b}{\partial z}\tfrac{\partial\varphi_a}{\partial z}\, d\Omega\Big]u^a_x$$
$$+\sum_a \Big[C_{1122}\int_\Omega \tfrac{\partial\varphi_b}{\partial x}\tfrac{\partial\varphi_a}{\partial y}\, d\Omega + \mu\int_\Omega \tfrac{\partial\varphi_b}{\partial y}\tfrac{\partial\varphi_a}{\partial x}\, d\Omega\Big]u^a_y$$
$$+\sum_a \Big[C_{1122}\int_\Omega \tfrac{\partial\varphi_b}{\partial x}\tfrac{\partial\varphi_a}{\partial z}\, d\Omega + \mu\int_\Omega \tfrac{\partial\varphi_b}{\partial z}\tfrac{\partial\varphi_a}{\partial x}\, d\Omega\Big]u^a_z = F_{bx}$$
$$\equiv \sum_a K^{11}_{ba}u^a_x + K^{12}_{ba}u^a_y + K^{13}_{ba}u^a_z = F_{bx} \qquad \texttt{[ElementStiffness.cs:122-168]}$$

7 — Матриця пружних сталих C та B-матриця

Матриця C (6×6, ізотропний матеріал):

$$\begin{pmatrix}\sigma_{xx}\\ \sigma_{yy}\\ \sigma_{zz}\\ \sigma_{xy}\\ \sigma_{yz}\\ \sigma_{xz}\end{pmatrix} = \underbrace{\begin{pmatrix} C_{1111} & C_{1122} & C_{1122} & 0 & 0 & 0\\ C_{1122} & C_{1111} & C_{1122} & 0 & 0 & 0\\ C_{1122} & C_{1122} & C_{1111} & 0 & 0 & 0\\ 0 & 0 & 0 & \mu & 0 & 0\\ 0 & 0 & 0 & 0 & \mu & 0\\ 0 & 0 & 0 & 0 & 0 & \mu \end{pmatrix}}_{C} \begin{pmatrix}\varepsilon_{xx}\\ \varepsilon_{yy}\\ \varepsilon_{zz}\\ \varepsilon_{xy}\\ \varepsilon_{yz}\\ \varepsilon_{xz}\end{pmatrix} \qquad \texttt{[ElementStiffness.cs:124-130]}$$

B-матриця (6×60): $\boldsymbol{\varepsilon} = B\mathbf{u}^{(e)}$, *DOF-порядок:* $[u_x^{0..19},\ u_y^{0..19},\ u_z^{0..19}]$
Стовпці 0..19 (блок B_x), 20..39 (B_y), 40..59 (B_z). Стовпець i кожного блоку:

$$B_x^{(i)} = \begin{pmatrix}\varphi_{ix}\\ 0\\ 0\\ \varphi_{iy}\\ 0\\ \varphi_{iz}\end{pmatrix}, \quad B_y^{(i)} = \begin{pmatrix}0\\ \varphi_{iy}\\ 0\\ \varphi_{ix}\\ \varphi_{iz}\\ 0\end{pmatrix}, \quad B_z^{(i)} = \begin{pmatrix}0\\ 0\\ \varphi_{iz}\\ 0\\ \varphi_{iy}\\ \varphi_{ix}\end{pmatrix}$$

$$K^{(e)} = \int_{-1}^{1}\int_{-1}^{1}\int_{-1}^{1} B^T C B\,|\det J|\,d\alpha\,d\beta\,d\gamma \approx \sum_k w_k\, B^{T(k)} C B^{(k)}\,|\det J^{(k)}| \qquad \texttt{[ElementStiffness.cs:36]}$$

Явне розкладання блоків через B_x, B_y, B_z:

$$K^{11} = \int B_x^T C_{11} B_x\,|\det J|\,d\Omega^{\text{ref}} \;\Rightarrow\; K_{ij}^{11} = \int (C_{1111}\varphi_{ix}\varphi_{jx} + \mu\varphi_{iy}\varphi_{jy} + \mu\varphi_{iz}\varphi_{jz})|\det J|\,d\Omega^{\text{ref}} \qquad \texttt{[ElementStiffness.cs:152]}$$

$$K^{12} = \int B_x^T C_{12} B_y\,|\det J|\,d\Omega^{\text{ref}} \;\Rightarrow\; K_{ij}^{12} = \int (C_{1122}\varphi_{ix}\varphi_{jy} + \mu\varphi_{iy}\varphi_{jx})|\det J|\,d\Omega^{\text{ref}} \qquad \texttt{[ElementStiffness.cs:161]}$$

Аналогічно $K^{13}, K^{22}, K^{23}, K^{33}$; симетрія: $K^{21}{=}(K^{12})^T$, $K^{31}{=}(K^{13})^T$, $K^{32}{=}(K^{23})^T$. [ElementStiffness.cs:182-186]

8 — Якобіан, обернення, глобальні похідні

9 компонент якобіана ($x^{(0)}{=}x, x^{(1)}{=}y, x^{(2)}{=}z$):

$$J_{00}{=}\sum_i \tfrac{\partial\varphi_i}{\partial\alpha}x_i,\ J_{01}{=}\sum_i \tfrac{\partial\varphi_i}{\partial\beta}x_i,\ J_{02}{=}\sum_i \tfrac{\partial\varphi_i}{\partial\gamma}x_i \qquad \texttt{[ElementStiffness.cs:94-96]}$$

$$J_{10}{=}\sum_i \tfrac{\partial\varphi_i}{\partial\alpha}y_i,\ J_{11}{=}\sum_i \tfrac{\partial\varphi_i}{\partial\beta}y_i,\ J_{12}{=}\sum_i \tfrac{\partial\varphi_i}{\partial\gamma}y_i \qquad \texttt{[ElementStiffness.cs:97-99]}$$

$$J_{20}{=}\sum_i \tfrac{\partial\varphi_i}{\partial\alpha}z_i,\ J_{21}{=}\sum_i \tfrac{\partial\varphi_i}{\partial\beta}z_i,\ J_{22}{=}\sum_i \tfrac{\partial\varphi_i}{\partial\gamma}z_i \qquad \texttt{[ElementStiffness.cs:100-102]}$$

9 кофакторів (c_{mn}):

$$c_{00} = J_{11}J_{22}{-}J_{12}J_{21},\quad c_{01} = J_{12}J_{20}{-}J_{10}J_{22},\quad c_{02} = J_{10}J_{21}{-}J_{11}J_{20} \qquad \texttt{[ElementStiffness.cs:202-204]}$$

$$c_{10} = J_{02}J_{21}{-}J_{01}J_{22},\quad c_{11} = J_{00}J_{22}{-}J_{02}J_{20},\quad c_{12} = J_{01}J_{20}{-}J_{00}J_{21} \qquad \texttt{[ElementStiffness.cs:205-207]}$$

$$c_{20} = J_{01}J_{12}{-}J_{02}J_{11},\quad c_{21} = J_{02}J_{10}{-}J_{00}J_{12},\quad c_{22} = J_{00}J_{11}{-}J_{01}J_{10} \qquad \texttt{[ElementStiffness.cs:208-210]}$$

$$\det J = J_{00}c_{00} + J_{01}c_{01} + J_{02}c_{02} \qquad \texttt{[ElementStiffness.cs:212]}$$

$$J^{-1} = \frac{1}{\det J}\begin{pmatrix} c_{00} & c_{10} & c_{20}\\ c_{01} & c_{11} & c_{21}\\ c_{02} & c_{12} & c_{22}\end{pmatrix} \qquad (J^{-1}_{mn} = c_{nm}/\det J) \qquad \texttt{[ElementStiffness.cs:214-224]}$$

Глобальні похідні (для кожного $i = 0..19$):

$$\frac{\partial\varphi_i}{\partial x} = J_{00}^{-1}\frac{\partial\varphi_i}{\partial\alpha} + J_{01}^{-1}\frac{\partial\varphi_i}{\partial\beta} + J_{02}^{-1}\frac{\partial\varphi_i}{\partial\gamma} \qquad \texttt{[ElementStiffness.cs:117]}$$

$$\frac{\partial\varphi_i}{\partial y} = J_{10}^{-1}\frac{\partial\varphi_i}{\partial\alpha} + J_{11}^{-1}\frac{\partial\varphi_i}{\partial\beta} + J_{12}^{-1}\frac{\partial\varphi_i}{\partial\gamma} \qquad \texttt{[ElementStiffness.cs:118]}$$

$$\frac{\partial\varphi_i}{\partial z} = J_{20}^{-1}\frac{\partial\varphi_i}{\partial\alpha} + J_{21}^{-1}\frac{\partial\varphi_i}{\partial\beta} + J_{22}^{-1}\frac{\partial\varphi_i}{\partial\gamma} \qquad \texttt{[ElementStiffness.cs:119]}$$

Спрощення для рівномірної прямокутної сітки ($h_s = A_s/(2N_s)$):

$$J = \begin{pmatrix} h_x & 0 & 0 \\ 0 & h_y & 0 \\ 0 & 0 & h_z \end{pmatrix}, \quad \det J = h_x h_y h_z = \text{const}, \quad J^{-1} = \begin{pmatrix} 1/h_x & 0 & 0 \\ 0 & 1/h_y & 0 \\ 0 & 0 & 1/h_z \end{pmatrix}$$

$$\frac{\partial \varphi_i}{\partial x} = \frac{1}{h_x}\frac{\partial \varphi_i}{\partial \alpha}, \quad \frac{\partial \varphi_i}{\partial y} = \frac{1}{h_y}\frac{\partial \varphi_i}{\partial \beta}, \quad \frac{\partial \varphi_i}{\partial z} = \frac{1}{h_z}\frac{\partial \varphi_i}{\partial \gamma}$$

9 — Вектор навантаження на грані

Карта вузлів кожної грані ($f = $ FaceId): [ElementLoad.cs:27-35]

Грань	Локальні вузли елемента (порядок: 4 кути + 4 середини)
Bottom (γ=−1)	0, 3, 2, 1, 11, 10, 9, 8
Top (γ=+1)	4, 5, 6, 7, 12, 13, 14, 15
Front (β=−1)	0, 1, 5, 4, 8, 17, 12, 16
Back (β=+1)	3, 7, 6, 2, 19, 14, 18, 10
Left (α=−1)	0, 4, 7, 3, 16, 15, 19, 11
Right (α=+1)	1, 2, 6, 5, 9, 18, 13, 17

Функції форми 8-вузлового серендіп-елемента грані:

$$\psi_i \ (i{=}0..3,\ \text{кут}) = \tfrac{1}{4}(1+\eta\eta_i)(1+\tau\tau_i)(\eta\eta_i+\tau\tau_i-1) \quad \text{[ShapeFunctions.cs:160-162]}$$

$$\psi_i \ (\eta_i{=}0) = \tfrac{1}{2}(1-\eta^2)(1+\tau\tau_i), \qquad \psi_i \ (\tau_i{=}0) = \tfrac{1}{2}(1+\eta\eta_i)(1-\tau^2) \quad \text{[ShapeFunctions.cs:168,170]}$$

Похідні функцій форми грані:

$$\frac{\partial\psi_i}{\partial\eta} \ (\text{кут}) = \tfrac{\eta_i}{4}(1+\tau\tau_i)(2\eta\eta_i+\tau\tau_i), \quad \frac{\partial\psi_i}{\partial\tau} \ (\text{кут}) = \tfrac{\tau_i}{4}(1+\eta\eta_i)(\eta\eta_i+2\tau\tau_i) \quad \text{[ShapeFunctions.cs:190-193]}$$

$$\frac{\partial\psi_i}{\partial\eta} \ (\eta_i{=}0) = -\eta(1+\tau\tau_i), \quad \frac{\partial\psi_i}{\partial\tau} \ (\eta_i{=}0) = \tfrac{\tau_i}{2}(1-\eta^2) \quad \text{[ShapeFunctions.cs:203-204]}$$

$$\frac{\partial\psi_i}{\partial\eta} \ (\tau_i{=}0) = \tfrac{\eta_i}{2}(1-\tau^2), \quad \frac{\partial\psi_i}{\partial\tau} \ (\tau_i{=}0) = -\tau(1+\eta\eta_i) \quad \text{[ShapeFunctions.cs:207-208]}$$

Дотичні вектори та площинна нормаль:

$$\frac{\partial x}{\partial\eta} = \sum_{i=0}^{7}\frac{\partial\psi_i}{\partial\eta}x_i^{(f)}, \quad \frac{\partial x}{\partial\tau} = \sum_{i=0}^{7}\frac{\partial\psi_i}{\partial\tau}x_i^{(f)}, \quad \text{аналогічно } y, z \quad \text{[ElementLoad.cs:78-84]}$$

$$N_x = \frac{\partial y}{\partial\eta}\frac{\partial z}{\partial\tau} - \frac{\partial z}{\partial\eta}\frac{\partial y}{\partial\tau}, \quad N_y = \frac{\partial z}{\partial\eta}\frac{\partial x}{\partial\tau} - \frac{\partial x}{\partial\eta}\frac{\partial z}{\partial\tau}, \quad N_z = \frac{\partial x}{\partial\eta}\frac{\partial y}{\partial\tau} - \frac{\partial y}{\partial\eta}\frac{\partial x}{\partial\tau} \quad \text{[ElementLoad.cs:87-89]}$$

$$F_{x,i}^{(e)} \mathrel{+}= w\,P_n\psi_i N_x, \quad F_{y,i}^{(e)} \mathrel{+}= w\,P_n\psi_i N_y, \quad F_{z,i}^{(e)} \mathrel{+}= w\,P_n\psi_i N_z \quad \text{[ElementLoad.cs:94-97]}$$

10 — Глобальне збирання

$$\text{dof}(c,n) = c\cdot N_{\text{nod}} + n, \quad c\in\{0{=}x, 1{=}y, 2{=}z\}, \quad n\in\{0,\dots,N_{\text{nod}}{-}1\} \quad \text{[GlobalAssembly.cs:116-117]}$$

$$p \ (\text{60-DOF}) \to \text{compP} = p/20,\ \text{lnP} = p\%20,\ \text{gdofP} = \text{compP}\cdot N_{\text{nod}} + \text{NT}[e, \text{lnP}] \quad \text{[GlobalAssembly.cs:115-117]}$$

$$n_g = \max_e\bigl(2N_{\text{nod}} + \max_{i\in e} n_i - \min_{i\in e} n_i\bigr) + 1 \quad \text{[GlobalAssembly.cs:48-52]}$$

$$\text{row} = \min(\text{gdofP}, \text{gdofQ}),\ \text{col} = \max(\text{gdofP}, \text{gdofQ}),\ \text{MG}[\text{row}, \text{col}{-}\text{row}] \mathrel{+}= K_{pq}^{(e)} \quad \text{[GlobalAssembly.cs:128-144]}$$

$$F[\text{gdofP}] \mathrel{+}= F_p^{(e)} \quad \text{[GlobalAssembly.cs:173]}$$

11 — Граничні умови (метод штрафу)

$$P = 10^{20}\cdot\max_i \text{MG}[i,0] \quad \text{[BoundaryConditions.cs:108]}$$

$$\text{MG}[\text{dof}, k] = 0 \ (k{\neq}0) \quad \text{— обнуляємо рядок} \quad \text{[BoundaryConditions.cs:114-115]}$$

$$\text{MG}[r,\ \text{dof}{-}r] = 0 \ (r{<}\text{dof}) \quad \text{— обнуляємо стовпець} \quad \text{[BoundaryConditions.cs:118-121]}$$

$$\text{MG}[\text{dof}, 0] = P, \quad F[\text{dof}] = 0 \quad \Rightarrow \quad P\cdot u_{\text{dof}} = 0 \ \Rightarrow\ u_{\text{dof}} = 0 \quad \text{[BoundaryConditions.cs:126-127]}$$

12 — Розв'язання СЛАР: метод Гауса на смуговій матриці

Пряме виключення ($k = 0 \to N{-}2,\ i = k{+}1 \to \min(k{+}n_g, N){-}1$):

$$m = \frac{\text{MG}[k,\ i{-}k]}{\text{MG}[k,0]} \quad \text{[GaussSolver.cs:48-49]}$$

$$F[i] \mathrel{-}= m\cdot F[k] \quad \text{[GaussSolver.cs:51]}$$

$$\text{MG}[i, j{-}i] \mathrel{-}= m\cdot\text{MG}[k, j{-}k], \quad j = i,\dots,\min(k{+}n_g, N){-}1 \quad \text{[GaussSolver.cs:55-62]}$$

Зворотній хід ($k = N-1 \to 0$):

$$F[k] \mathrel{-}= \sum_{j=k+1}^{\min(k+n_g,N)-1} \mathrm{MG}[k, j-k] \cdot F[j] \qquad \texttt{[GaussSolver.cs:81-85]}$$

$$u[k] = F[k] \mathrel{/}= \mathrm{MG}[k, 0] \qquad \texttt{[GaussSolver.cs:87-91]}$$

13 — Постпроцесинг

Витяг переміщень з глобального вектора U:

$$u_x^{(i)} = U[0 \cdot N_{\mathrm{nod}} + \mathrm{NT}[e,i]], \quad u_y^{(i)} = U[1 \cdot N_{\mathrm{nod}} + \mathrm{NT}[e,i]], \quad u_z^{(i)} = U[2 \cdot N_{\mathrm{nod}} + \mathrm{NT}[e,i]] \quad \texttt{[StressComputation.cs:95-98]}$$

Усі 6 компонент тензора деформацій у вузлі (через $\frac{\partial \varphi_i}{\partial x/y/z}$ з розд. 8):

$$\varepsilon_{xx} = \sum_{i=0}^{19} u_x^{(i)} \frac{\partial \varphi_i}{\partial x} \qquad \texttt{[StressComputation.cs:147]}$$

$$\varepsilon_{yy} = \sum_{i=0}^{19} u_y^{(i)} \frac{\partial \varphi_i}{\partial y} \qquad \texttt{[StressComputation.cs:148]}$$

$$\varepsilon_{zz} = \sum_{i=0}^{19} u_z^{(i)} \frac{\partial \varphi_i}{\partial z} \qquad \texttt{[StressComputation.cs:149]}$$

$$\varepsilon_{xy} = \sum_{i=0}^{19} u_x^{(i)} \frac{\partial \varphi_i}{\partial y} + \sum_{i=0}^{19} u_y^{(i)} \frac{\partial \varphi_i}{\partial x} \qquad \texttt{[StressComputation.cs:151]}$$

$$\varepsilon_{yz} = \sum_{i=0}^{19} u_y^{(i)} \frac{\partial \varphi_i}{\partial z} + \sum_{i=0}^{19} u_z^{(i)} \frac{\partial \varphi_i}{\partial y} \qquad \texttt{[StressComputation.cs:153]}$$

$$\varepsilon_{xz} = \sum_{i=0}^{19} u_x^{(i)} \frac{\partial \varphi_i}{\partial z} + \sum_{i=0}^{19} u_z^{(i)} \frac{\partial \varphi_i}{\partial x} \qquad \texttt{[StressComputation.cs:155]}$$

Закон Гука (σ через ε):

$$\sigma_{xx} = C_{1111}\varepsilon_{xx} + C_{1122}(\varepsilon_{yy} + \varepsilon_{zz}) \qquad \texttt{[StressComputation.cs:160]}$$

$$\sigma_{yy} = C_{1111}\varepsilon_{yy} + C_{1122}(\varepsilon_{xx} + \varepsilon_{zz}) \qquad \texttt{[StressComputation.cs:161]}$$

$$\sigma_{zz} = C_{1111}\varepsilon_{zz} + C_{1122}(\varepsilon_{xx} + \varepsilon_{yy}) \qquad \texttt{[StressComputation.cs:162]}$$

$$\sigma_{xy} = \mu\varepsilon_{xy}, \quad \sigma_{yz} = \mu\varepsilon_{yz}, \quad \sigma_{xz} = \mu\varepsilon_{xz} \qquad \texttt{[StressComputation.cs:163-165]}$$

Осереднення по вузлу n:

$$\sigma_{\alpha\beta}^{(n)} = \frac{1}{\mathrm{cnt}_n} \sum_{e \ni n} \sigma_{\alpha\beta}^{(e,n)}, \qquad \mathrm{cnt}_n = \#\{e : \mathrm{NT}[e,\cdot] = n\} \qquad \texttt{[StressComputation.cs:183-191]}$$

Напруження Мізеса:

$$\sigma_{\mathrm{VM}} = \sqrt{\tfrac{1}{2}\left[(\sigma_{xx}-\sigma_{yy})^2 + (\sigma_{yy}-\sigma_{zz})^2 + (\sigma_{zz}-\sigma_{xx})^2 + 6(\sigma_{xy}^2+\sigma_{yz}^2+\sigma_{xz}^2)\right]} \qquad \texttt{[StressComputation.cs:201-204]}$$

N_{nod}=`nqp`; $\mathrm{MG}[r,0]=K_{rr}$ (діагональ); $\mathrm{MG}[r,s-r]=K_{rs}$ для $s \geq r$ (верхній трикутник); $F[\cdot]$ після розв'язання $\equiv u[\cdot]$.